SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 1 – Interactive Tree Simulator 100
Weightage:	30% of CIA	**Total Marks:**	
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.*		
Student Name:	P. Harsha vardhan	**Reg. No.:**	192411179
Section / Batch:	CSE	**Date:**	27/07/2026

Code of Conduct

I P. Harsha Vardhan (192411179) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: p harsha

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
	GRAND TOTAL:	**100 Marks**	

Annexure A – Interactive Tree Simulator

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1. Construct a Binary Search Tree (BST) by inserting the following keys: 40, 20, 10, 30, 60, 50, 70
   Perform:
   a. Inorder Traversal
   b. Search for 70
   c. Delete 50
   d. Display the final BST.
2. Complete Tree Traversals by Inserting 55,35,75,25,45,65,85. Find all the traversals. a. Inorder b. Preorder
   c. Postorder

3. Insert 40, 20, 60, 10, 30, 50, 70, 5 into an AVL Tree. Determine the balance factor of every node before and after balancing.
4. Construct a B-Tree of order 3 by inserting 10, 20, 30, 40, 50.
5. Construct a Red-Black Tree using 50, 40, 60, 30, 45, 55, 70, 20. Delete 30 and analyze how Red-Black properties are restored.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

Question 1:

SORTING

GRAPH

DATA STRUCTURES

Circular Queue

Linked List

DOubly Linked List

Binary Heap

Binary Search Tree

AVL Tree

Red—Black Tree

AVL Tree vs Red —Black Tree

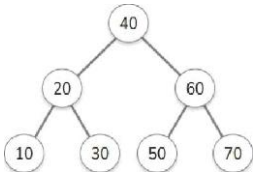
Splay*ree

Deletion first locates the target node, then carefully reconnects its children so the BST rule still holds. If the node has no children, it is simply removed. I+ it has one child, that child takes its place. If the node has two children, the tree replaces it with a nearby node that preserves ordering. Learn the full strategy in our BST deletion guide.

Pseudocode

```
function insert(node, key):
    if key ‹ node .vaLre
        if node.left is null:
            node.left = new Node(key)
        eLse:
            insert(node.left, key)
    else if key > node.value:
        if node.right is null:
            node.right = new Node(key)
        else:
            insert(node. right, key)
```

Enter a number: 81 INSERT | | DELETE | | CLEAR | | C B



*-BockToHome

Tree Troversols

Con trols

30 20 60 40 50 10 70

a0

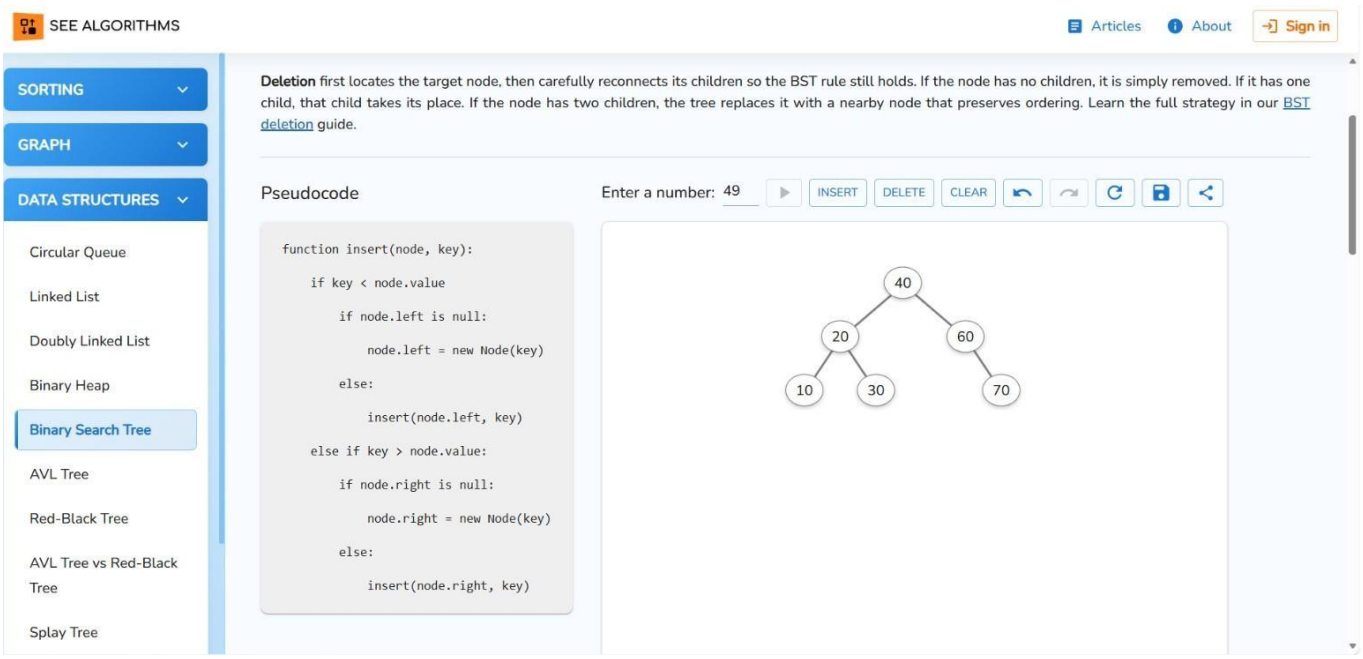
20 10

30 60 50 70

Pre +n Post

¥• \f isunlizo

¥' start Simulation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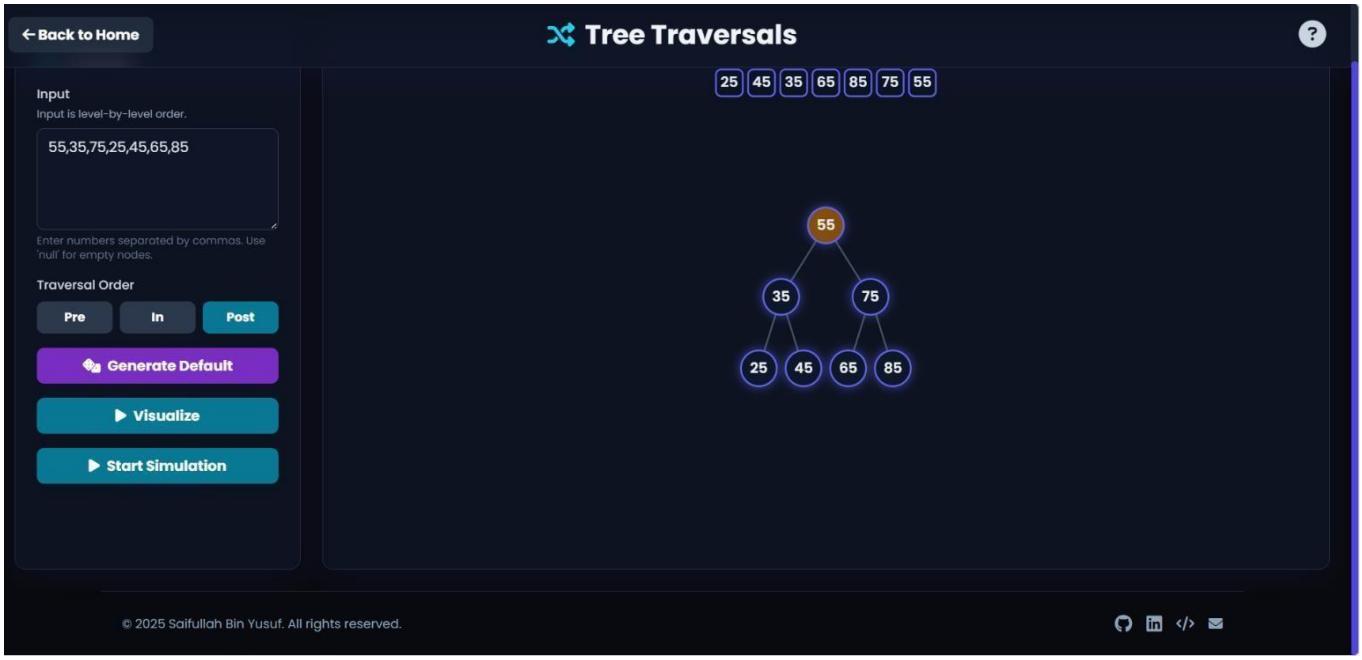
DATA STRUCTURES
Circular Queue
Linked List
Doubly Linked List
Binary Heap
Binary Search Tree
AVL Tree
Red-Black Tree
AVL Tree vs Red-Black Tree
Splay Tree
Deletion first locates the target node, then carefully reconnects its children so the BST rule still holds. If the node has no children, it is simply removed. If it has one child, that child takes its place. If the node has two children, the tree replaces it with a nearby node that preserves ordering. Learn the full strategy in our BST deletion guide.
Pseudocode
Enter a number: 49
INSERT
DELETE
CLEAR
function insert(node, key):
if key < node.value
if node.left is null:
node.left = new Node(key)
else:
insert(node.left, key)
else if key > node.value:
if node.right is null:
node.right = new Node(key)
else:
insert(node.right, key)
40
20
60
10
30
70

Question 2:


← Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
55,35,75,25,45,65,85
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
55 35 25 45 75 65 85
55
35
75
25
45
65
85


← Back to Home
Tree Traversals
Input
Input is level-by-level order.
55,35,75,25,45,65,85
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
25 45 35 65 85 75 55
55
35
75
25
45
65
85
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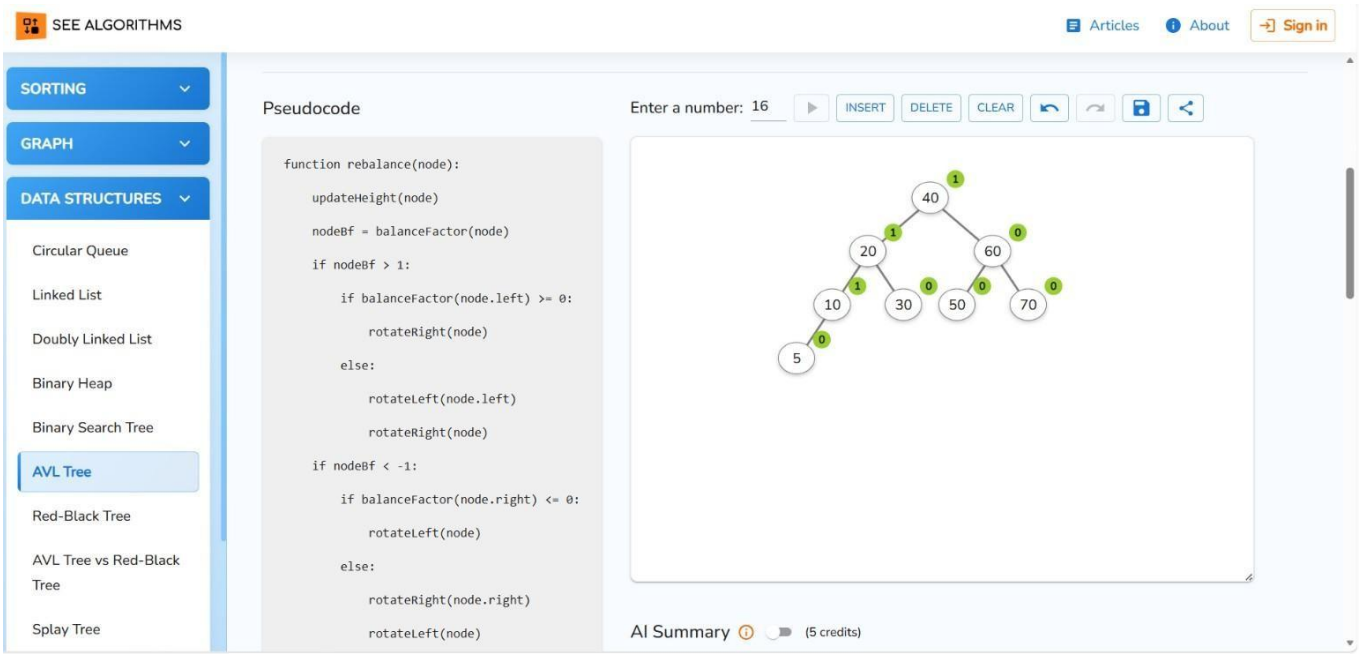
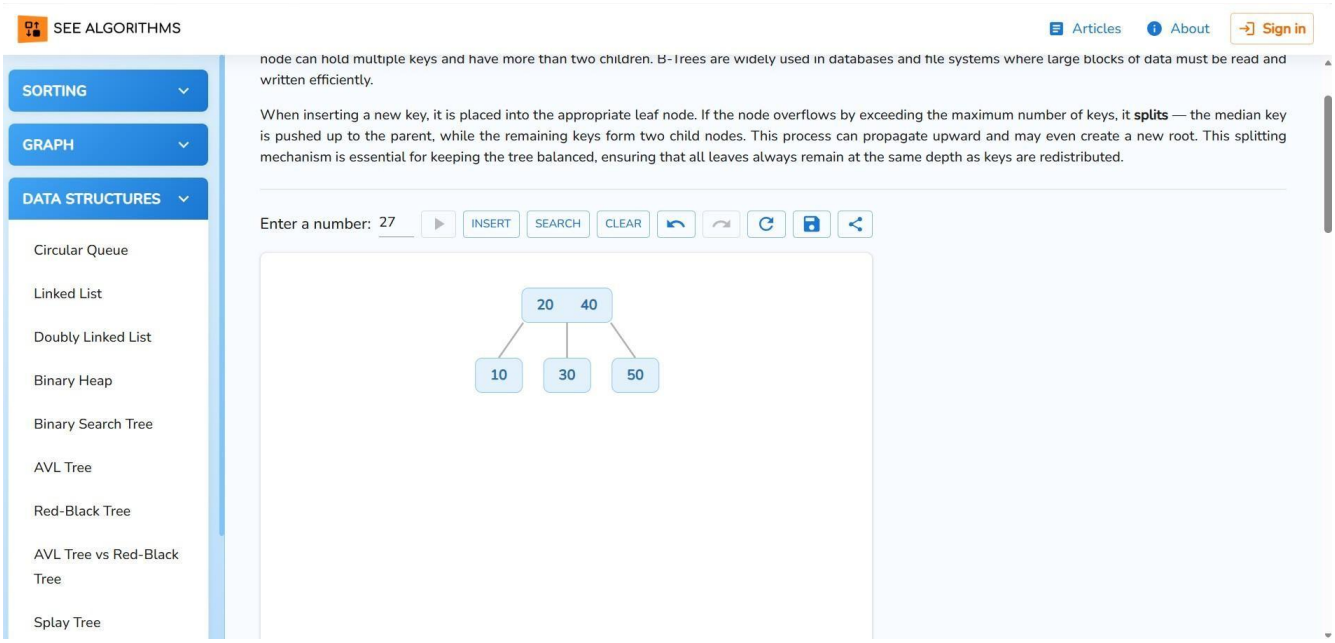
Question 3:



Question 4:



Question 5:

SORTING

GRAPH

DATA STRUCTURES

Doubly Linked List

Binary Heap

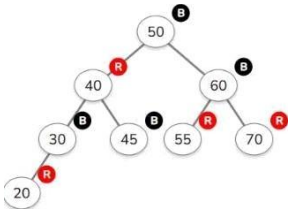
Binary Search *ree

AvAe

AVL Tree vs Red-Black Tree

rotations. It maintains batance through four key properties: the root is always black, all teaves snuff nodes} are black, red nodes @rinot has red children, and every path from a node to its descendant leaves contains the same number of bba¢k nodes.

Enter a number: * iNsER! cLE°R -



3. If there's a viotafion, apply rotations hell or right| and recotoring to restore properties.
4. Repeat urrtil aft rules are sahisfled from fhe modified node fa The root.